Design and Performance of Conveyor Belt using PID for Industrial Applications

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***Abstract*—One of the main motors utilized in robotic system applications are DC motor. In contrast to brushless DC motors, which are difficult to regulate, a brushed permanent magnet DC motor will be explored in this study because input voltage may be adjusted to revolve in either a clockwise or counterclockwise direction. Building a state-space model of a brushed permanent magnet DC motor using Simulink and integrating the plant with a PID controller by adjusting the PID settings to obtain the desired response are the outcomes of the study. Modeling the electro-mechanical system's equations was the initial stage. The approach section will describe these models and the associated equations. The state-space form will be produced from this model in order to create function blocks. These function blocks will be created and organized in Simulink using an open-loop method. The system will test the plant while simulating and analyzing the waveforms of the current, angular velocity, and angular displacement. In the case of a closed-loop system, the intended answer was set to 1, and the PID settings (Controller) had to be altered to ensure that the actual response obtained after running through the plant model was 1.**

Keywords—Conveyor Belt, PID, Closed-Loop, Open-Loop, State Space Model, Transfer Function

I. Introduction

A conveyor is a mechanical system capable of transporting products between locations. It has been extensively used in the business for the transportation of huge and environmentally-friendly commodities. During the manufacturing process, the weight of the goods varies considerably. If a conveyor is overloaded, it might affect the motor/conveyor speed if it is not equipped with speed control.

In many areas of human endeavor, belt drives are utilized to transfer mechanical energy from a spinning shaft to the control objects. In our daily lives, belt drives are used in a wide variety of products, including computers, automobiles, and audio and video equipment. In the industrial setting, these drives can be utilized for movement (like conveyors) or object control placement (like the exact positioning of semiconductor components on circuit plates). In the past, belt systems have been utilized to transport electricity between rotating machine components with up to 98% efficiency. Conveyors are a typical use for such a system; the earliest prototype has been in use since the 19th century. Additionally, Henry Ford built the first assembly line using conveyor belts at the Ford Motor Company's plant as early as 1913. Today, belt drive systems, are widely used in the automobile sector and in conveyor-type applications [1].

As the load on the conveyor grows, the anchor current will also increase. The increase in current will amplify the interference of the anchor field in the main field, resulting in a continual displacement of the neutral point. If the location of the brush is not altered, shifting the primary field will increase losses on the brush.

In research done by Stephanus Antonius Ananda, the average efficiency of a motor without speed control is roughly 70%, which is quite low. Using a PID controller, researchers have developed a motor speed control system. In the research, a line-following robot is used as a plant with a maximum overrun of 10%. The output rate will be affected. The output rate will be affected. A PID controller is one of the various motor speed control methods made possible by scientific and technological advancements. Using DC motors, a number of past works have investigated speed control. Mohammad and Billah did a study analysis on speed control of DC series motors using a diverter and noting sites of speed saturation. Using a diverter in the motor's field circuit, the speed of a dc series motor may be adjusted [2]. Sahoo investigated the linear quadratic regulator and predictive control model used for efficient speed regulation of a DC motor. In the DC motor model, the speed angle is controlled using multiple control approaches [3].

Dil Kumar investigates the dynamic SMC control scheme using a PID controller that is fitted to the adaptive speed control of a DC motor. Separate speed control for the excited DC motor is presented using adaptive PID and dynamic shear mode control approaches [4] Das did research on the best tuning of the PID controller using the GWO algorithm to regulate the speed of the DC motor. Methods of bio-inspired meta-heuristic soft computing to improve the performance of the PID controller utilized in DC motor speed control [5] Jeddi's study into solar energy facilitates the speed regulation of independently powered DC motors. Controlling the speed of the direct current motor (SEDM) independently utilizing the Maximum Power Point Tracking (MPPT) approach used to the DC-DC Boost converter [6]. Saqib explored the linearization technique for anti-windup synthesis and application in DC motor speed control. Anti-windup gain is intended for nonlinear systems with input saturation [7]. Neural Network-based study was undertaken by Nath on energy-efficient DC drive speed control. DC motors provide superior speed control properties and are extensively used in a variety of industrial applications [8].

Sun examined the design and simulation of a Proteus-based DC PWM motor speed regulator. The Proteus program is used to develop and simulate PWM speed regulators of DC motors and to get the motor's primary control [9]. Ravi Kumar researched PWM-based microprocessor-based closed-loop control of DC motor speed.

The majority of electric drive systems must possess these defining characteristics, including linear control, dependability, and steady operation [10]. Jaziri studied remote DC motor control utilizing Embedded Linux and FPGA. [11] The Codeign design in which the embedded Linux operating system running on Beaglebone interfaces with the FPGA via the SPI protocol and a master-slave method.

Chaudhary researched the speed regulation of DC motors using ANFIS. Comparative examination of several controller designs for controlling the speed of DC motors [12] Bhatti explored the comparison of P-I and I-P controllers utilizing the Ziegler-Nichols tuning technique for speed control of DC motors. The Ziegler-Nichols tuning technique for controlling the speed of a DC motor using a Proportional-Integral (P-I) and Integral-Proportional (I-P) controller [13]. Sai researched the control of the speed of a DC motor using voice instructions. There are several methods for controlling speed, such as utilizing sound as an input to eliminate manual operation [14]. The modeling of a brushless DC motor speed control system using Simulink was investigated. Ling Xu's brushless mathematical model, according to his study. MATLAB/Simulink is used to construct a number of distinct functional modules for DC motors, which are then merged into a simulation model of a DC motor control system [15]. Patil examines the control speed of a four-quadrant closed DC motor.

Control with little cost and excellent performance Cutter-Based Four DC motors with closed loops [16] Yurkevich researched PWM speed control of DC motors using single perturbation methods. [16] The proportional-integral (PI) controller is intended to manage the motor current and speed based on a single perturbation approach so as to artificially produce scale-time movement in a closed loop system. Nguyen Thanh explored the improvement of the performance of the unfiltered DC motor control utilizing fuzzy logic. Despite the introduction of other kinds of electric motors, DC motors continue to be utilized in a variety of industrial applications because they are easy to regulate and have a large range of adjustment [17].

II. Methods

In this section, the function blocks which will be put in the Simulink were derived from the standard equations:

$$V_{in} = L\frac{di}{dt} + Ri + k_e\omega \tag{1}$$

Where: Vin is the input voltage, L is the motor inductance, R is the resistance, ke is the back emf constant, ω is the angular velocity, and i is the motor current.

The Motion Equation was introduced:

$$J\ddot{\theta} + c\dot{\theta} = T_m - T_{load} \tag{2}$$

Where: J is the equivalent inertia, c is the equivalent viscous damping, Tm is the motor input torque, $Tload$ is the other load from the system, $\ddot{\theta}$ is the angular acceleration, and $\dot{\theta}$ is the angular velocity.

In this report, the J formula is:

$$J = J_{motor} + \frac{J_{robot}}{G^2} \tag{3}$$

Where: J_{motor} is the motor inertia, J_{robot} is the robot inertia, and G is the gear ratio.

From equation (2), T_m can be substituted with $k_m i$

$$T_m = k_m i \tag{4}$$

Where: k_m is the motor torque constant and i is the motor current.

Combines System Model Equation was introduced:

From equation (1), (2), and (3), the electrical and dynamic relationships were illustrated by the equations below:

$$J\ddot{\theta} + c\dot{\theta} - k_m i = -T_{load} \tag{5}$$

$$L\frac{di}{dt} + Ri + k_e\dot{\theta} = V_{in} \tag{6}$$

State-Space Form Equations were introduced:

From equations (5) and (6), the state space equations need to be formed by considering the 3 output variables (Angular displacement (θ), angular velocity ($\dot{\theta}$), and motor current (i)). Each of the variable ($x_1 = \theta$, $x_2 = \dot{\theta}$, $x_3 = i$) was substituted from equations (5), (6) and will result in first order ODE.

$$\dot{x}_1 = x_2 \tag{7}$$

$$\dot{x}_2 = -\frac{c}{J}x_2 + \frac{k_m}{J}x_3 - T_{load} \tag{8}$$

$$\dot{x}_3 = -\frac{k_e}{L}x_2 - \frac{R}{L}x_3 + \frac{V_{in}}{L} \tag{9}$$

Block-Diagram Form equation was introduced: T_{load} from equation (8) was neglected since the simulation will not take input from any external load. Next, the state-space form equations were transformed to Laplace equations and then transformed to block diagram equations.

$$sx_1 = x_2$$

$$\frac{x_1}{x_2} = \frac{1}{s} \tag{7}$$

$$sx_2 = -\frac{c}{J}x_2 + \frac{k_m}{J}x_3 - T_{load}$$

$$\frac{x_2}{x_3} = \frac{k_m}{Js+c} \tag{8}$$

$$sx_3 = -\frac{k_e}{L}x_2 - \frac{R}{L}x_3 + \frac{V_{in}}{L}$$

$$\frac{x_3}{V_{in}-k_e x_2} = \frac{1}{Ls+R} \tag{9}$$

These three equations (7,8,9) were transformed to block diagrams and were arranged similar with the figure below. The left-hand side of each of the equation is the output (numerator) and input (denominator).

III. System Desing & Implementation

From the equation generated from the above section are all modelled using MATLAB 2022a Simulink software.

A. Open-Loop Model using Simulink

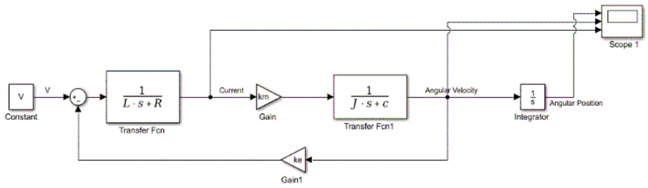

Fig. 1 Open Loop System of DC Motor for the conveyor belt system

In figure 1, the initiation of our discussion from the left side of the diagram. A constant block is the square piece that has been given the designation "V." The block shown in this schematic was responsible for supplying the plant with an input voltage signal of 12V. The sum block may be identified by its round form, which is immediately to the right of the plus and minus signs (it can add or subtract input). In this instance, the gain of the "Gain1" is considered the negative input, while the positive input is the 12V voltage signal. The result of this calculation will be sent to a location known as the "Transfer Fcn."

A gain block with the name "Gain1" can be seen at the bottom of the diagram. This gain block will double the value coming from the "Transfer Fcn1" block, and the result will be sent to the sum block. The rear emf constant value is maintained by this gain block (ke). In the upper right-hand corner, we can make out the letters "Transfer Fcn." Since this is the block of the transfer function that is connected to equation 9, the output will be current. The form of a triangle may be seen out to the right. The term "Gain" refers to a gain block that multiplies the value that is being transferred from "Transfer Fcn" to "Transfer Fcn1."

A constant number for motor torque is included in this gain block (km). The phrase "Transfer Fcn1" is shown over on the right-hand side. The output is the angular velocity since this is the transfer function block that is associated with equation 8. On the right hand side, there is a block that resembles an integrator, and it is connected to equation 7. The purpose of this block is to do the calculation necessary to determine the value of the integral of the signal that it is fed as a function of time. The angular position is the result of this block's processing in this instance. In addition, it is clear from looking at figure 3 that the analysis will be performed on three scopes: the current scope, the angular velocity scope, and the angular position scope.

B. Closed-Loop Model using Simulink

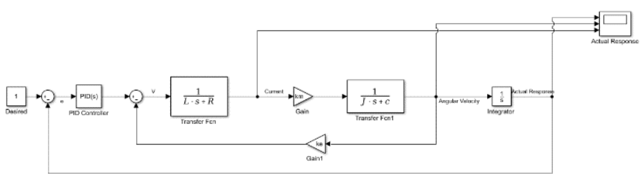

Fig. 2 Closed Loop System of DC Motor with PID for the conveyor belt system

The input voltage signal for the second section (Closed Loop System of DC Motor with PID for the conveyor belt system) was adjusted to a constant value of 1. The sum block will then be provided the actual response value. This block will add the positive value of 1 with the actual response's negative value. The outcome of this computation was sent to the sum block of the prior open loop system via PID controller.

IV. Results & Analysis

A. Open-Loop System

i. Without changing the payload mass (5 kg) and the motor inertia (1.3 × 10−4 kgm^2)

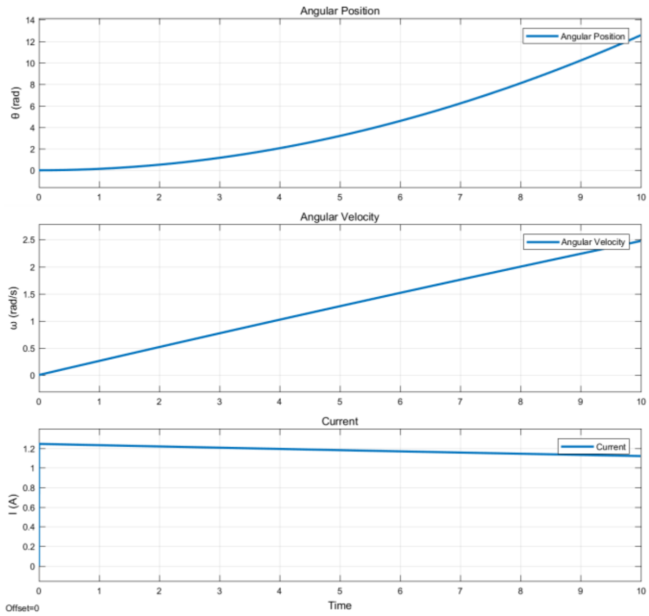

Fig. 2 Waveforms of Angular Position, Angular velocity, and Current for 10 seconds

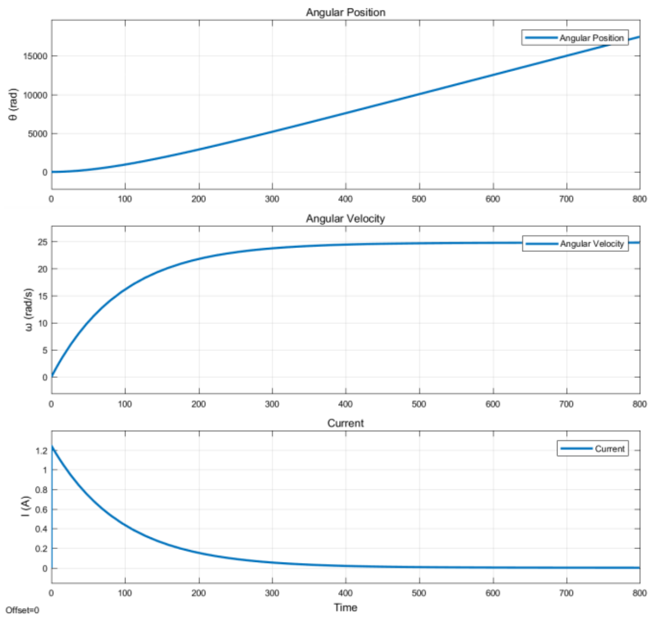

Fig. 3 Waveforms of Angular Position, Angular velocity, and Current for 800 seconds

The waveforms of the angular location, angular velocity, and current are shown in Figure 2. Initially, we can observe that the current peaked at 1.243A at 0.043 seconds and fell to 1.12A at 9.9 seconds. If we take the mean of these two figures, we have a current of 1.18A. For the first 10 seconds, the angular velocity graph was linear, reaching 2.5 rad/s at time 10 seconds. To get the maximum angular velocity, we have to prolong the time period to determine what would occur to the graph. Figure 3 clearly demonstrates that the velocity rose from zero to sixteen radians per second in 100 seconds. However, it began to approach saturation below 25 rad/s. At 780 seconds, the speed achieved its maximum value of 24.78 rad/s, and the current value was virtually zero, 0.0009851A, according to the cursor measurement. This was due to the torque and speed connection shown in Figure 4. If there were no load, the motor could reach this maximum speed. Due to equation 4 in the technique section, the motor torque was proportionate to the current under this situation of no load. Therefore, the motor can hardly handle load at 24.78 rad/s. In addition, the maximum power occurs when the speed is half of the maximum speed, therefore it occurs at about 12.39 rad/s.

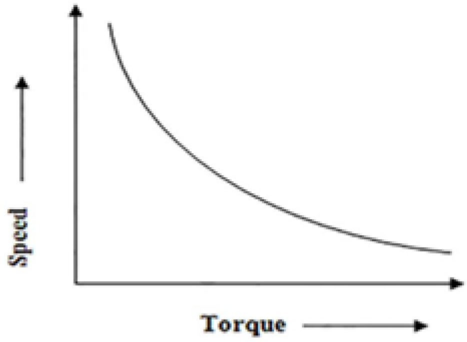

Fig. 4 Torque-Speed Relationship

The value of the angular displacement rose gradually at first and then linearly as a function of time. The angular displacement reached about 12.5 radians at time 10second. Figure 3 shows that the displacement likewise rose linearly with time, reaching around 17000 radians after 800 seconds.

ii. Changing the payload mass to (10kg) and the motor inertia to $(4.3 \times 10-4\,kgm^2)$

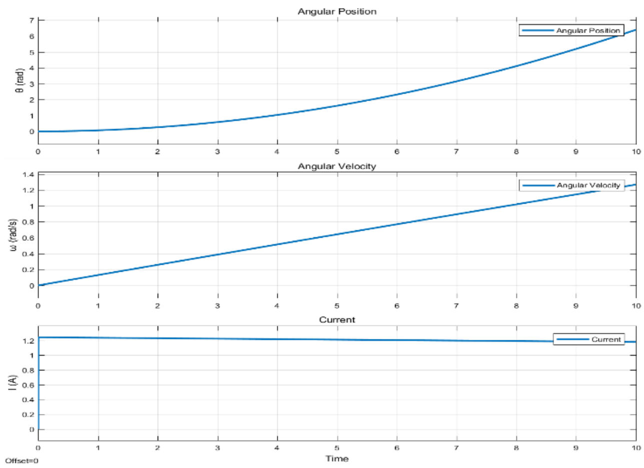

Fig. 5 Waveforms of Angular Position, Angular velocity, and Current for 10 seconds with different mass and inertia

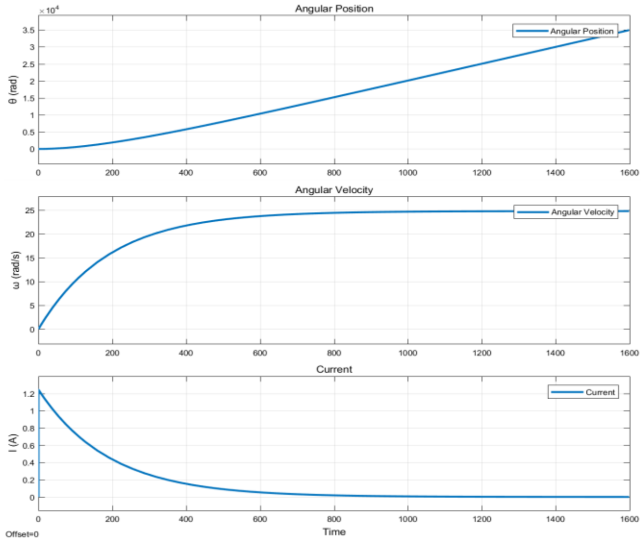

Fig. 6 Waveforms of Angular Position, Angular velocity, and Current for 1600 seconds with different mass and inertia

Figure 5 illustrates current, angle, and velocity waveforms. First, the peak current was 1.243A at 0.043 seconds and 1.18 at 9.9 seconds. This averages out at 1.15A.

The angular velocity graph was linear for 10 seconds, peaking at 1.3 rad/s. We had to increase the time period to determine the maximum angular velocity. Figure 6 illustrates the initial velocity increase from 0 to 7.36 rad/s in 100 seconds. Below 25 rad/s, saturation commenced. The cursor speed reached 24.78 rad/s at 1540 seconds, when the readout was 0.001023. Maximum power was also 12.39 rad/s because the maximum speed was the same. Angular displacement increased gradually before linearly with time. At 10 seconds, angular displacement was 6.4 radians. Figure 6 shows that the angle grew linearly during 800 seconds, reaching 15000 radians.

B. Closed-Loop System

i. Without changing the payload mass (5kg) and the motor inertia $(1.3 \times 10-4\,kgm^2)$

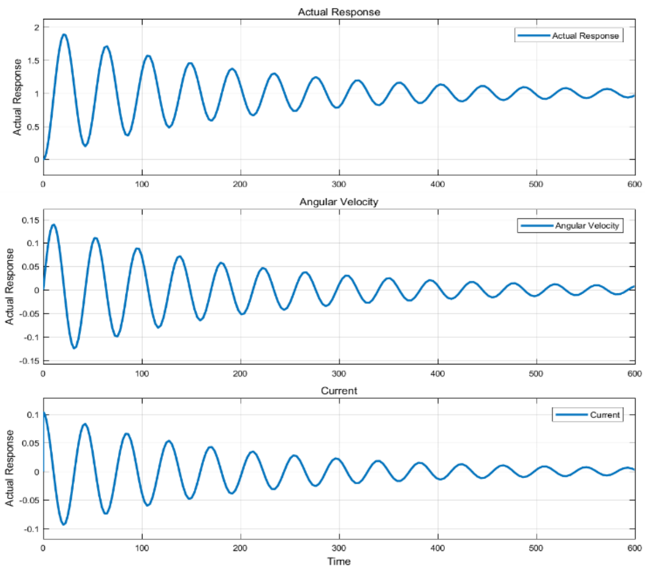

Fig. 7 Waveforms of Angular displacement (actual response), Angular Velocity, and Current without PID block

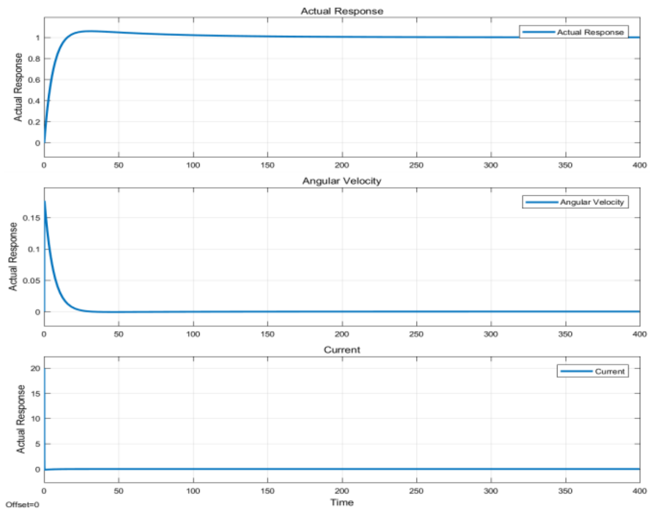

Fig. 8 Waveforms of Angular displacement (actual response), Angular Velocity, and Current with PID value from V.

Figure 7 showed that the simulation's genuine reaction, angular velocity, and current oscillated (600 second). Three plots showed oscillation amplitudes falling after starting high. The graphs demonstrate that the genuine response reference was 1, but 0 for angular velocity and current. Rise time and peak overshoot were investigated for reaction. Peak overshoot is the most the response exceeded the desired value in the first oscillation, whereas rise time is the time it took for the response to grow from 10% to 90% of the final value [18, 19]. Cursor measurements found that the increasing time was roughly 10 seconds, and the peak overshoot was 0.889. Figure 8 showed that the real response, angular velocity, and current waveforms first varied significantly, were oscillation-free, and subsequently returned to their reference values until the simulation's end (400 second). By focusing on the actual reaction, we can see that the graph had a minimum peak overshoot of 0.058, a rising time of 10.829 seconds, and no steady state error. This answer is 1 when 350 seconds pass. The PID settings were confirmed because they delivered the anticipated response (a stable close loop system) while preserving a short rise time (approximately 10 seconds), low peak overshoot, and no steady-state error or oscillation.

ii. Changing the payload mass to $(10kg)$ and the motor inertia to $(4.3 \times 10-4kgm^2)$

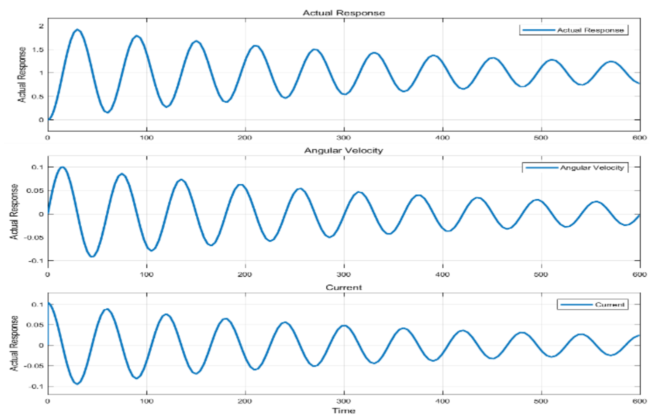

Fig. 9 Waveforms of Angular displacement (actual response), Angular Velocity, and Current without PID block (With increase of payload + inertia)

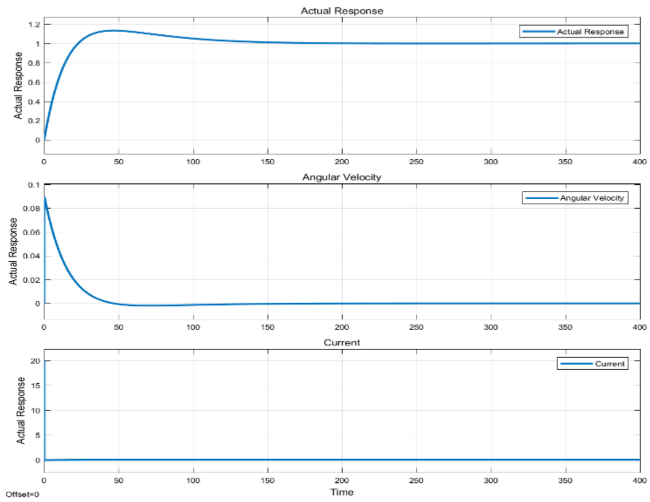

Fig. 11 Waveforms of Angular displacement (actual response), Angular Velocity, and Current with PID value from V. (With increase of payload + inertia)

Figure 10 made it very evident that the waveforms of the real response, angular velocity, and current oscillated throughout the simulation (600 second). These three graphs demonstrated that oscillation amplitudes were dropping and that they were beginning with a high value. The figures show that the reference response for the real response was 1, and that it was 0 for both angular velocity and current. The rising time and peak overshoot were examined with an eye toward the real reaction. The rising time was 14.330 seconds, and the peak overshoot was 0.923 seconds, according to the cursor readings.

Figure 11 made it evident that the waveforms of the real response, angular velocity, and current underwent significant variations at first, were oscillatory-free, and then started to return to their reference values until the simulation's conclusion (400 second). With a clear focus on the actual response, it is possible to observe that the graph was a perfect damping graph with a negligibly small peak overshoot of 0.118, a rising time of 19.4 seconds, and no steady state inaccuracy. When the duration is about 700 seconds, this response will precisely equal 1. (More than the time given in figure 11). The PID settings were therefore confirmed since they produced the intended response (stable close loop system), which nevertheless maintained the quick rising time (about 10 second), minimal amount of peak overshoot, removal of steady state error, and avoidance of excessive oscillation.

V. SETTING UP THE PID PARAMETERS

i. Without changing the payload mass $(5kg)$ and the motor inertia $(1.3 \times 10-4kgm^2)$

The PID values that were chosen for the close loop system with original payload mass and inertia (figure 12) was Kp = 0.21, Ki = 0.0012, and Kd = 8.32 where Kp is the proportional, Ki is the integral, and Kd is the derivatives.

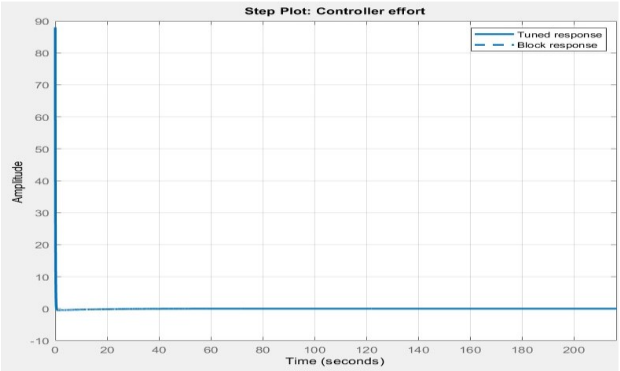

Fig. 12 Controller effort for PID for Close Loop System with Original payload mass and inertia

	Tuned	Block
Rise time	65.3 seconds	9.88 seconds
Settling time	604 seconds	103 seconds
Overshoot	12.7 %	5.85 %
Peak	1.13	1.06
Gain margin	100 dB @ 20.8 rad/s	80.9 dB @ 141 rad/s
Phase margin	69 deg @ 0.0212 rad/s	84.3 deg @ 0.182 rad/s
Closed-loop stability	Stable	Stable

Fig. 13 Chosen PID Value vs Recommended PID Value (For system with original payload mass and

Before choosing these PID Values for the closed loop system, they underwent comprehensive analysis. The controller effort was impacted by the PID values that were selected. The controller will have to work harder as a result, as evidenced by the fact that it attained an amplitude of around 88 (Figure 12). The suggested PID Values provide controller effort with an amplitude smaller than 1, but they also have a long rising time, settling time, and overshoot. Because the DC motor plant needed a faster rise time, the "Chosen PID" was chosen. The close loop system was likewise far from being unstable because the gain and phase margin values were both far from being negative.

ii. Changing the payload mass to (10 kg) and the motor inertia to ($4.3 \times 10{-}4 kgm^2$)

The PID values that were chosen for the close loop system with original payload mass and inertia (figure 12) was Kp = 0.173, Ki = 0.00077, and Kd = 7.819 where Kp is the proportional, Ki is the integral, and Kd is the derivatives

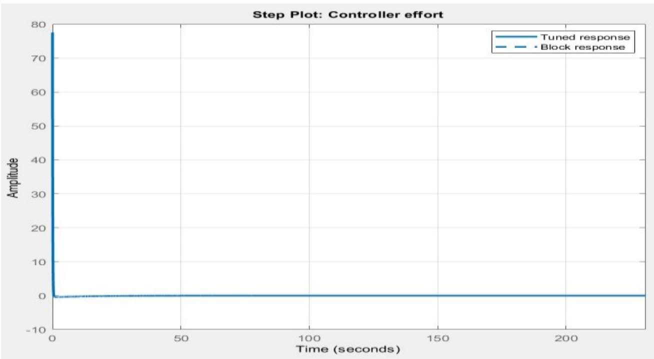

Fig. 14 Controller effort for PID for Close Loop System with different payload mass and inertia

	Tuned	Block
Rise time	142 seconds	18 seconds
Settling time	1.55e+03 seconds	147 seconds
Overshoot	10.5 %	11.8 %
Peak	1.1	1.12
Gain margin	106 dB @ 8.14 rad/s	87.5 dB @ 141 rad/s
Phase margin	69 deg @ 0.00923 rad/s	78.5 deg @ 0.0866 rad...
Closed-loop stability	Stable	Stable

Fig.15 Chosen PID Value vs Recommended PID Value (For system with different payload mass and inertia)

These PID Values were selected thoroughly before sending them for the close loop system. The selected PID values had an impact for the controller effort. This will make the controller to work harder, it was proved that it achieved amplitude of around 77 (Figure 17). Although the recommended PID Values offer controller effort with amplitude that less than 1, it had significant rise time and settling time compared to the “Chosen PID”. Thus, the “Chosen PID” was selected since a quicker rise time was required for the DC motor plant. In addition, both gain and phase margin values were also far away from negative and these values indicated that the close loop system was far away from instability.

VI. Discussion & Conclusion

It was possible to verify the current using these two open loop systems since it reached its peak at 1.24A and maintained a current of around 1.1A throughout. When it came to the angular velocity, the first open loop system achieved 16 rad/s within the first 100 seconds, but the second one (which had increased mass and inertia) reached 7.36 rad/s. In addition, the maximum value of the angular velocity, which is 24.78 rad/s, will be reached at 780 seconds, but the other system would require a greater amount of time (1540 seconds). Because of this, the angular velocity will decrease as the payload mass and inertia continue to grow. In addition, the current that was flowing through either of the open loop systems at 780 or 1540 seconds was very close to zero. This confirmed that the connection between torque and speed predicts that the maximum speed would result in zero torque, and will therefore result in zero current. The first open loop system obtained 17000 radians for the angular displacement at 800 seconds, whereas the other system with the larger load reached 15000 radians. It is possible to draw the conclusion that a greater moment of inertia as well as the mass of the cargo will cause the angular displacement to travel a shorter distance. Therefore, the angular velocity is proportional to the angular displacement; nevertheless, both of these quantities have an inverse relationship with the mass of the payload and the inertia of the motor.

The results of these two closed-loop systems without PID control made it abundantly evident that an increase in payload mass and inertia would cause the system oscillation to become larger and last for a longer period of time. As a consequence, the rise time was longer, and the overshoot was larger. Due to the fact that these two closed loop systems did not provide the correct answer, a controller was added to the close loop system in order to generate the needed response while also removing any mistakes that may have been there.

The Proportional-Integral-Derivative (PID) control algorithm was chosen for this closed-loop system because it offered more flexibility than any other controller, including the Proportional-Integral controller. The derivative control that was also required to cope with motion control, such as a DC motor system, prevented the adoption of the PI algorithm [20]. In addition, the capacity of the PID Controller to recover from disturbances was shown to be superior to that of the P and PI Controller.

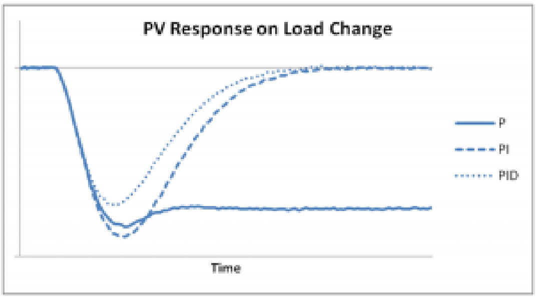

Fig. 16 Comparison between P, PI, and PID Controller on Ability to Recover from Disturbance

The PID value for the first close loop system with the original payload and inertia was Kp = 0.21, Ki = 0.0012, and Kd = 8.32, while the PID value for the second close loop system, which had a somewhat lower value, was Kp = 0.173, Ki = 0.00077, and Kd = 7.819. Because each of the values has an influence on the rising time, overshoot, settling time, and steady state error [21] each and every one of them was picked with great care. The proportional control was tuned to minimize the rise time, the derivative control was tuned to decrease the overshoot, and the integral control was tuned to decrease the steady state error. This was a condensed version of the process. It can be seen in figures 12 and 14 that both of the intended responses were accomplished by achieving a rapid rising time, minimizing peak overshoot, and eliminating steady-state errors. In addition, both the phase

CL RESPONSE	RISE TIME	OVERSHOOT	SETTLING TIME
Kp	Decrease	Increase	Small Change
Ki	Decrease	Increase	Increase
Kd	Small Change	Decrease	Decrease

margin and the gain margin pointed to the system being stable. As a result, the PID values for the closed loop system of the DC Motor were examined and confirmed.

VII. Reference

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